

$$\begin{array}{ccc}
 F([X, Y]_{\mathcal{C}}) & \xrightarrow{\varphi_{X, Y}} & [FX, FY]_{\mathcal{D}} \\
 \downarrow \alpha_{[X, Y]_{\mathcal{C}}} & & \downarrow [FX, \alpha_Y]_{\mathcal{D}} \\
 & & [FX, GY]_{\mathcal{D}} \\
 & & \uparrow [\alpha_X, GY]_{\mathcal{D}} \\
 G([X, Y]_{\mathcal{C}}) & \xrightarrow{\psi_{X, Y}} & [GX, GY]_{\mathcal{D}}
 \end{array}$$